Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

Sector	Field content
even	$[1,1]$, $[3,3]$, $[5,5]$, $[7,7]$, $[9,9]$, $[11,11]$, $[13,13]$
odd	$[2,2]$, $[4,4]$, $[6,6]$, $[8,8]$, $[10,10]$, $[12,12]$
twisted	$[1,13]$, $[3,11]$, $[5,9]$, $[7,7]$, $[9,5]$, $[11,3]$, $[13,1]$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{c} M_H L_2(p) = E_t(p) L_1(p) - \hat{I}_q \left\{ \left[c_p^{(-)} c_q^{(-)} + \hat{p}\cdot\hat{q} s_p^{(-)} s_q^{(-)} \right] L_1(q) \right\} \\ M_H L_1(p) = E_t(p) L_2(p) - \hat{I}_q \left\{ \left[c_p^{(+)} c_q^{(+)} + \hat{p}\cdot\hat{q} s_p^{(+)} s_q^{(+)} \right] L_2(q) \right\} \end{array} \right. .$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$D = \left(\begin{array}{cc} v^{-1}\partial_1 & u^{-1}\partial_2 \\ u\partial_4 & v\partial_3 \end{array} \right) , \quad \bar{D} = \hat{V}^{\epsilon\,-1} D = \left(\begin{array}{cc} v\partial_3 & -q^{-2}u^{-1}\partial_2 \\ -q^{-2}u\partial_4 & v^{-1}\partial_1 \end{array} \right) ,$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{1}{\beta_L}\mathbb{E}\Big(G^\Lambda(z_L;n,n)\Big) &\leq \frac{1}{Imz}\,\pi\Big(1+\frac{k}{2}\Big)S_\mu\Big(\frac{2\ Imz_L}{k}\Big), \quad \Lambda = C_p,\ \Lambda_L \\ &\leq C\,\left(\frac{2\ \beta_L^{-1}\ Imz}{k}\right)^\alpha \\ &\leq C\ (2L+1)^{-d}, \quad (\text{since}\ \ \beta_L = (2L+1)^{d/\alpha}). \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{rcl} T^* & = & \mathcal{D}T\mathcal{D}^{-1} \\ & = & \left(\begin{array}{ccccc} t_{11} & t_{12} & -t_{13} & t_{14} & t_{15} \\ t_{21} & t_{22} & -t_{23} & t_{24} & t_{25} \\ -t_{31} & -t_{32} & t_{33} & -t_{34} & -t_{35} \\ t_{41} & t_{42} & -t_{43} & t_{44} & t_{45} \\ t_{51} & t_{52} & -t_{53} & t_{54} & t_{55} \end{array} \right) . \end{array}$$